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碩士論文

基於 CycleGAN 和特徵融合於挑戰性場景的車道偵測
Lane Detection Based on CycleGAN and Feature Fusion in
Challenging Scenes

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中文摘要

車道偵測是智能駕駛系統的關鍵技術，透過辨識車道的位置和形狀，使車輛能夠保持在正確的車道上並避免事故。基於影像的深度學習方法實現車道偵測是現在最先進的方法，在一般的白天場景已經有相當出色的辨識能力，並且幾乎能夠達到實時偵測。然而，這些模型經常在夜晚、炫光或陰影等挑戰性的場景中無法準確的辨識車道。此外，訓練數據有限的多樣性限制了模型處理不同環境的能力。這篇論文提出了一個新穎的方法，通過現有的白天和夜間的資料集訓練 CycleGAN，我們的方法得以萃取不同風格及多尺度的特徵，從而增加模型輸入的豐富性。我們使用 CycleGAN 作為領域適應模型結合影像分割模型，以增強模型在不同風格場景的性能。所提出的一致性損失函數來減少模型在不同場景中的性能差異。實驗結果表明，我們的方法提高了原始車道偵測模型在挑戰性場景中的偵測表現。這項研究有助於提高智能駕駛系統的可靠性和穩健性，最終使道路更安全並增強駕駛體驗。

關鍵字：車道偵測、影像風格轉換、特徵融合、挑戰性場景

Abstract

Lane detection is a crucial technology of the intelligent driving system. By identifying the position and shape of the lane, the vehicle can stay in the correct lane and avoid accidents. The image-based deep learning method for lane detection is currently the most advanced method. It already has a very good recognition ability in general daytime scenes, and can almost achieve real-time detection. However, these models often fail to accurately identify lanes in challenging scenarios such as night, dazzle, or shadows. Furthermore, the lack of diversity in the training data limits the ability of the model to handle different environments. This thesis proposes a novel method to train CycleGAN with existing daytime and nighttime datasets. Our method can extract features of different styles and multi-scales, thereby increasing the richness of model input. We use CycleGAN as a domain adaptation model combined with an image segmentation model to improve the performance of the model in different styles of scenes. The proposed consistent loss function is used to reduce the performance variance of the model in different scenarios. The result of the experiment shows that our method improves the detection performance of original lane detection models in challenging scenarios. This research helps improve the dependability and robustness of intelligent driving systems, ultimately making roads safer and enhancing the driving experience.

Keywords: lane detection, image style transfer, feature fusion, challenging scenes

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Chapter 1 Introduction

1.1 Background

According to the 2022 report of Markets & Markets [22], an international research and survey organization, the market value of self-driving artificial intelligence may reach US\$7 billion in 2027. Improving driving safety and efficiency, reducing accidents, and enhancing the driving experience have been long-term goals in the development of intelligent driving systems such as self-driving vehicles and advanced driver assistance systems (ADAS). Autonomous driving technology includes many key assistance systems, including lane centering assist system (LKAS) and lane departure warning system (LDWS). The goal of LKAS is to assist the vehicle to keep in the center of the lane. The system uses lane detection technology to identify lane lines and monitor the vehicle's position. When the vehicle deviates from the center of the lane, it achieves stability and comfort through slight automatic adjustments, helping to reduce fatigue driving and the risk of accidents. LDWS also monitors the vehicle's position and lane lines through lane detection technology. When the vehicle deviates from the lane, the system will sound, vibrate or visually alert the driver to remind the driver to take action. Such systems can help drivers avoid accidents, especially lane departures due to distraction or fatigue.

Lane detection is the core technology of these systems. Lane detection uses sensors and cameras to capture lane markings and features on the road and uses image processing technology to identify the position and shape of the lane. This provides critical information for the intelligent driving system to ensure that the vehicle stays in the correct lane and avoids drifting or colliding with other vehicles. In recent years, lane detection methods have been significantly improved due to image-based deep learning research. The deep learning model can learn how to extract the features of the lane based on the dataset to achieve high-accuracy

lane detection. This method solves the requirement that the manual feature extraction method is difficult to deal with the complex and variable road environment, and improves the accuracy and robustness of the system. The image-based deep learning lane detection technology enables self-driving technologies such as LKAS and LDWS to be realized, and the driving experience becomes safer and more comfortable.

1.2 Motivation

In recent years, lane detection methods based on deep learning models have achieved outstanding performance and can almost achieve real-time detection. Although these models perform well in general daytime scenarios, their performance degrades significantly at night or in other challenging scenarios. Low-light conditions and insufficient lighting at night, as well as other factors such as strong glare, shadows, rain, or snow, can impact the performance of the lane detection system. At night, images suffer from insufficient external lighting, noise, motion blur, and optical blur, and lane markings also have reduced contrast, making them harder to identify. In other challenging scenarios, such as shadowed or wet roads, the clarity and visibility of lane lines can also suffer, making detection difficult.

Another important reason is that the diversity of the training dataset is insufficient. The deep learning model used for lane detection relies on a large amount of training data, which is used to update the parameters of the model and improve the ability of the model to recognize the characteristics of the lane. However, when the training data lacks enough samples of challenging scenes, the performance of lane detection models in these scenarios will be limited. Collecting and labeling data for scenes that include various weather conditions or otherwise is costly in terms of labor and time. In recent years, there have been many advanced researches on domain adaptation models based on generative adversarial networks, which are used to solve the problem of data distribution differences between different domains. Image style transfer is one of the important applications, which involves

the technique of transferring the style of one image to another. SIM-CycleGAN [20] uses a style transfer method to convert daytime images into nighttime images to augment the training dataset for lanes. However, such a data augmentation method has limited improvement in the performance of the model, and cannot reduce the difference in the detection ability of the model in different scenarios. We believe that while doing style conversion, we can simultaneously consider extracting multi-scale features and constraining the output differences of different scenes. This inspired us to use the domain adaptation model to do image style conversion and combine feature extraction methods to solve the issue of insufficient diversity of training data.

1.3 Research approach

To address these challenges, CycleGAN [37] is a promising solution that can transform images from one domain to another without relying on labeled data. CycleGAN is a generative adversarial network (GAN) that has numerous applications in domain adaptation tasks, such as converting ordinary photos into artwork or game footage into real-world environments. Unlike other style transfer methods, CycleGAN provides a more flexible solution for real-world applications since it does not require strictly paired images. This is an indispensable advantage in our method since it is almost impossible to collect paired day and night images in road scenes. Our proposed method will use the existing day and night images of the lane dataset to train CycleGAN, then use the trained CycleGAN to extract different styles and multi-scale features and combine image segmentation models to enhance the detection ability of the model in different styles of scenes. Experiments prove that our method can improve the detection performance of the original lane detection model in various scenarios.

1.4 Contribution

1. The proposed method will combine the domain adaptation model and image segmentation model to improve the detection performance of the model in challenging scenes.
2. Use the domain adaptation model to extract different styles and multi-scale feature training models to improve the robustness of the model.
3. The proposed consistency loss function constrains the final output of feature maps of different styles in the model, which will reduce the performance difference of the model in different scenarios.
4. Through experimental research to understand the impact of various methods and training strategies on the lane detection model.
5. In the public lane test dataset, our method can enhance the F1 score of the original model by 1% to 3.3%.

1.5 Organization

The organization of this thesis is described in the section that follows. In Chapter 2, we review the research on domain adaptation, lane detection, and domain adaptation combined with image recognition. Chapter 3 explains our proposed method and model architecture. Chapter 4 presents the experimental results and analysis of our proposed method. Chapter 5 presents the conclusion and future work of the thesis.

Chapter 2 Related Work

The literature review explores two main areas: domain adaptation and lane detection. In the domain adaptation section, it discusses the use of GAN to transfer information between different image domains. The lane detection section focuses on deep learning methods for detecting lane lines in road scenes. Lastly, the combination of domain adaptation and image recognition is discussed for enhancing performance in challenging road scene recognition tasks.

2.1 Domain Adaptation

In recent years, GAN has become a popular and effective method for learning and generating realistic data distributions. In computer vision, image domain adaptation is a challenging task that seeks to transfer information learned from an original domain to a target domain where labeled data is scarce or unavailable. Traditional approaches often rely on handcrafted features or explicit alignment methods, which may not generalize well to diverse and complex image distributions. However, with the emergence of GAN, novel techniques have been introduced to tackle image domain adaptation by learning mappings between different image domains. CoGAN [21] is a notable generative model that extends the concept of traditional GAN to learn a joint distribution of multiple domains simultaneously. It uses the same noisy input to generate two styles of images. It ensures that the overall appearance of the two outputs is similar by sharing the shallow weights of the generator. pix2pix [12] proposes a method based on conditional GAN to realize image-to-image conversion and uses UNet [26] as the network architecture of the Generator. This method is limited by the need for pairs of images to train the model.

BicycleGAN [38] mainly consists of two different GANs. The encoder maps real images to their corresponding latent vectors, while the decoder reconstructs images from the

latent vectors. Unlike most one-to-one transformation models, their method can produce diverse results. UNIT [19] presents a novel approach that combines GAN and variational autoencoders to achieve high-quality image translation without the need for paired examples. To achieve this, UNIT employs two main components: a shared-latent space and domain-based encoders and decoders. The shared-latent space serves as a bottleneck layer, where both the content and style representations are encoded. Images from one domain are mapped to the shared-latent space and vice versa by domain-based encoders and decoders. CycleGAN [37] has significantly advanced the field of image domain adaptation by enabling unpaired image-to-image translation. A cycle-consistency loss is introduced, requiring that translated images from the target domain be mapped back to their originating domain while maintaining their content. A novel viewpoint for image domain adaptation is provided by DualGAN [33], which expands upon the idea of unpaired image-to-image translation. It introduces a reconstruction loss, the concept of which is the same as CycleGAN cycle-consistency loss. This loss enforces the translated images to remain close to the original images in both the original and target domains, enhancing the preservation of content and style during the adaptation process. DiscoGAN [13] uses the concept of learning bidirectional mappings between domains to provide a novel approach for unpaired image-to-image translation. The DiscoGAN method shares similarities with CycleGAN. It focuses primarily on small-sized images in its experimental analysis. Additionally, DiscoGAN introduces a distinct model architecture that sets it apart from CycleGAN.

MUNIT[11] proposes a polymorphic unsupervised image translation framework, assuming the image representations can be divided into content feature vectors and style feature vectors. By separating these two components, MUNIT enables independent manipulation of content and style, allowing for the generation of diverse translations without the need for paired training data. DRIT[15] uses two encoders to respectively output the

context space that retains the image content and a domain-based attribute space. As long as the attribute space of different images is used as input, images of different domains can be generated. INIT [29] is based on the architecture of MUNIT and divides the domain transformation into global style and instance style to guide the model to generate more complete objects. Although paired images are not required, images with bounding boxes are required for training. CIGAN [23] focuses on unsupervised low-light enhancement methods, and it also uses low-light images and normal-light images to train the model. It adds noise in the generator to simulate the noise of low-light images and synthesizes more realistic low-light images.

The comparison in Table 1 of papers in this section is based on (1) the data requirement is paired images or unpaired images ; (2) the mapping relation is one to one or one to multiple; (3) The main contributions are reconstruction constraints, shared latent weights, or adding noise.

Table 1. Related works of domain adaption methods.

	Data Requirement		Mapping Relations		Main Contribution		
	Paired Images	Unpaired Images	One-to-one	One-to-multiple	Reconstruction Constraint	Shared Latent	Adding Noise
CoGAN		✓	✓			✓	✓
pix2pix	✓		✓				✓
BicycleGAN	✓			✓	✓		✓
UNIT		✓	✓		✓	✓	
CycleGAN		✓	✓		✓		
DualGAN		✓	✓		✓		
DiscoGAN		✓	✓		✓		
MUNIT		✓		✓	✓		
DRIT		✓		✓	✓		
INIT		✓	✓		✓		
CIGAN		✓	✓		✓		✓

2.2 Lane Detection

Lane detection is essential to the visual navigation system of intelligent vehicles, and it has always been one of the popular area of research in autonomous driving. In recent years, lane detection methods based on deep learning models have achieved remarkable performance and can almost achieve real-time detection. Table 3 summarizes the recent comparisons of lane detection methods. Wang *et al.* propose a deep learning lane line detection method called LaneNet [32], which demonstrates high stability in both highway and urban road without relying the number of lanes. Liang *et al.* propose a convolutional neural network (CNN) architecture called LineNet [16] with lane prediction layers and scaling modules. SCNN [24] improves the traditional CNN layers into sequential

convolutional layers to facilitate the information transfer between pixels in rows and columns, thereby enhancing lane feature extraction. EL-GAN [5] is the first method to use a GAN for lane detection. This network simultaneously uses the predicted lane results and label-trained discriminator to propose an embedding loss to stabilize the entire network's training and achieve better lane prediction results.

ENet-SAD [9] introduces a new self-attention distillation method that enables the model to learn from feature output maps at different scales, capturing rich contextual information. This lightweight model achieves advanced lane detection performance and decent detection speed. ConvLSTM [39] proposes a model that combines CNN and recurrent neural networks and introduces a new method for lane detection using consecutive images. Compared to single-image-based methods, using multiple images can effectively detect lanes in complex scenes with shadows or occlusions. LaneATT [30] proposes an anchor-based lane detection model. Lane distribution often follows specific patterns, and anchor detection based on global information can generate more accurate lane positions even when lane markings are occlusion or absent. Ripple-GAN [35] utilizes a Wasserstein generative adversarial network and semantic segmentation to learn lane line features. By classifying targets into the background, lane surface, and lane lines, it exhibits better detection capability for complex or ambiguous lane lines. LaneAF [1] predicts binary segmentation and generates corresponding lane instances by processing horizontal and vertical clustering row by row. This method demonstrates efficient detection capability without requiring a predefined maximum number of lanes.

PINet [14] uses a multi-loss function trained hourglass model. The clustering problem of predicting critical points is regarded as an instance segmentation problem, and the model size can be chosen based on the target's computational capacity. The model can be trained regardless of the number of lane lines. BezierLaneNet [4] introduces a novel method for

defining lane lines based on curve parameters. Utilizing parameterized Bezier curves solves the optimization difficulties encountered in traditional polynomial curve representation of lane lines during model training. Additionally, it possesses advantages such as ease of computation and high stability. The model can maintain high accuracy while operating with low latency. CLRNet [36] focuses on exploiting high-level semantic features and low-level texture features in lane detection. By utilizing more contextual information, it can obtain local lane features to improve localization accuracy. SCNN_Unet_LSTM [3] combines the lane feature extraction method of SCNN and the method of ConvLSTM that uses multiple images for lane prediction to enhance detection capability in challenging driving scenarios.

Table 2 shows (1) the main improvement part about feature extraction, loss function, or post-processing; (2) Whether to use supervised learning or unsupervised learning; (3) A single image or a continuous image is utilized as the input of the model.; (4) the output of the model is pixels or points.

Table 2. Related works of lane detection methods.

	Main Improvement			Fundamental Approach		Input Type		Output Type	
	Feature Extraction	Loss Function	Post-processing	Supervised	Unsupervised	Single Image	Image Sequence	Pixel	Point
LaneNet	✓			✓		✓		✓	
LineNet	✓			✓		✓		✓	
SCNN	✓			✓		✓		✓	
EL-GAN		✓			✓	✓		✓	
ENet-SAD	✓			✓		✓		✓	
ConvLSTM	✓			✓			✓	✓	
LaneATT	✓	✓	✓	✓		✓			✓
Ripple-GAN		✓			✓	✓		✓	
LaneAF		✓	✓	✓		✓		✓	
PINet		✓		✓		✓			✓
BezierLaneNet	✓	✓		✓		✓			✓
CLRNet	✓	✓		✓		✓			✓
SCNN_Unet_LSTM	✓			✓			✓	✓	

2.3 The Combination of Domain Adaption and Image Recognition

For image recognition of road scenes, including recognition of vehicles, street signs or lane lines, etc. Collecting training data is hard work that costs a lot of time and money. Most of the existing road datasets lack challenging scenes, making deep learning models unable to perform stably in these challenging scenes. How to use the method of domain adaptation to improve the issue of insufficient training data has also become potential research. CyCADA [7] converts the image to the target domain and will input the translated image

into the semantic segmentation model to calculate the loss of semantic segmentation, and also calculate the semantic consistency loss with the original image. This model successfully achieves good performance in both road environment scenes and digit classification.

AugGAN [10] also uses the domain adaptation method to generate night images to expand the data set of detected vehicles. It uses an encoder paired with two decoders as a model architecture. One of the decoders outputs the synthesized image, and the other outputs the image segmentation map. This constrains the model to maintain the integrity of the object when generating synthetic images. SIM-CycleGAN [20] proposes a data enhancement method based on style conversion, which converts daytime images into nighttime images through CycleGAN, thereby increasing the proportion of nighttime images. M-AugGAN [18] is based on the method of multimodal GAN. In a model architecture similar to AugGAN, random vectors are input to achieve various output results, to generate diverse and realistic night images to assist model training. DUNIT [2] first uses the detection subnetwork to detect instance objects and then processes the image and object in parallel. After fusing the two, it uses the decoder to generate the full image. Generated images can strike a balance between keeping objects and translating images.

FTEGanOd [28] uses the CycleGAN feature fusion enhancement module to cascade the object detection module to improve vehicle detection performance in complex lighting environments. It builds an end-to-end network combining an image domain adaptation network and object detection network and uses a feature fusion enhancement module to combine night and day features to enhance night vehicle detection. In order to preserve detected objects during image translation, AugGAN-Det [17] adds the object detection model to the GAN model for joint training. The generated image is guided to maintain the integrity of the object as much as possible so that the generated image is more conducive to expanding the dataset of object detection. Most approaches treat the image as a whole and

directly input it into the model, which makes it difficult to preserve the objects in the image well or keep them real.

To compare the research on the combination of domain adaptation and image recognition, we made the following Table 3, comparing (1) the type of image recognition is semantic segmentation or object detection; (2) the research method is aimed at data enhancement or image translation; (3) The utilization of two-stage training, instance consistency, or feature fusion is the main contribution of the method.

Table 3. Related works of the combination of domain adaption and image recognition.

	Detection Type		Objective		Main Contribution		
	Semantic Segmentation	Object Detection	Data Enhancement	Image Translation	Two-stage Training	Instance Consistency	Feature Fusion
CyCADA	✓			✓		✓	
AugGAN	✓		✓			✓	
SIM-CycleGAN	✓		✓		✓		
M-AugGAN	✓		✓			✓	
DUNIT		✓		✓		✓	✓
FTEGanOd		✓	✓		✓		✓
AugGAN-Det		✓	✓			✓	

Chapter 3 Methodology

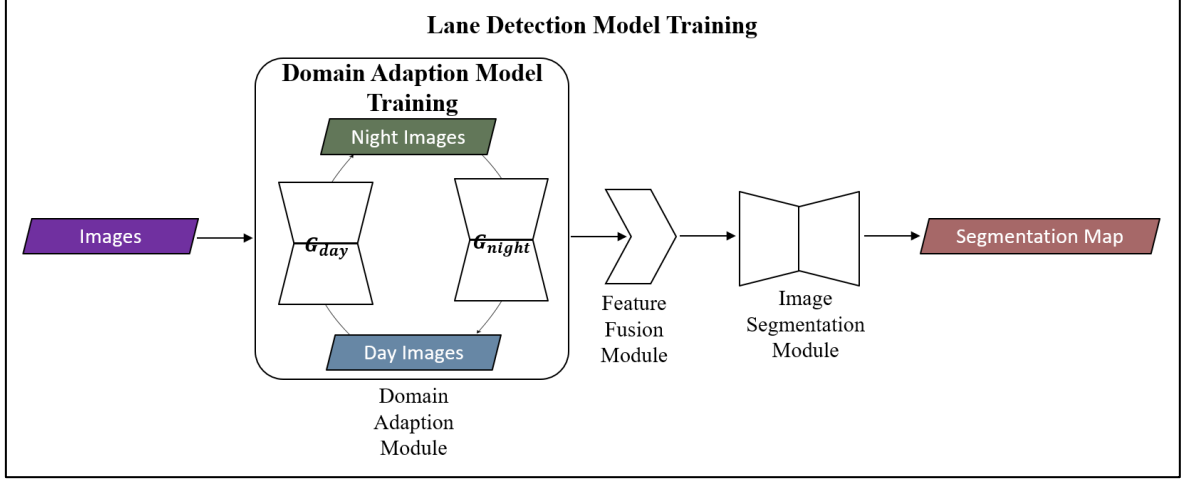


Figure 1. Illustration of the two-stage proposed method.

In order to better extract the lane features of challenging scenes, the domain adaptation model is employed to extract various styles of image features, which are then utilized as input for the image segmentation model in lane detection. The framework of our proposed method, which is primarily separated into two stages, is depicted in Figure 1. The first stage is the training of the domain adaptation model, and the second stage is the training of the lane detection model. Using day and night images, the CycleGAN model is trained in the first stage of domain adaption model training, and the trained CycleGAN Generator parameters are saved. The domain adaption module, the feature fusion module, and the image segmentation module are combined in the second stage of the lane detection model for lane detection training. In Section 3.1, we introduce CycleGAN and describe how it is used to train a domain adaptation model that transforms night and day images. In Section 3.2, we will introduce the domain adaptation module, feature fusion module, and image segmentation module in the lane detection model respectively. We also detail our proposed consistency loss function and the entire model training process.

3.1 Domain Adaption Model Training

For training the position of the lane, the majority of current lane detection models

simply input images into deep learning models, so they are limited by the richness and diversity of the training dataset. Most of the image data in existing lane datasets are daytime scenes. In general daytime and sunny scenes, deep learning models often have excellent recognition performance. But in the face of relatively rare scenes, such as night, rainy, foggy, and other scenes. Due to the limitation of the dataset, the model cannot have the same recognition performance in these scenarios. The lane detection model we propose includes a domain adaption module, a feature fusion module, and an image segmentation module. The image will be input into the domain adaptation module first, and the feature fusion module will output the feature maps representing the Encoder and Decoder of the domain adaption module, and then enter the image segmentation module together for training and testing. Since our domain adaptation model is a model that converts day and night images into each other, the feature maps output by the Encoder and Decoder can just represent the characteristics of day or night. That is to say, our lane detection model can use one image to train feature maps representing day and night at the same time. Therefore, our method can enhance the lane detection ability of the model at night and in other challenging scenes with different lighting conditions.

Based on the above motivation, we need to first train a domain adaptation model that can convert night and day images to each other. We use CycleGAN as our domain adaptation model. The CycleGAN model is an unsupervised image translation method. Its objective is to learn to map between two domains without paired training data. Not requiring paired data is an important advantage of using CycleGAN, since it is almost impossible to collect paired day and night road scenes in our approach. However, paired labels are not required to use CycleGAN, only night and day images are required. CycleGAN differs from other image translation models by its ability to perform image translation in both directions, called cycle consistency. The model is capable of converting images from domain X to domain Y while

maintaining the properties of the original image. The model can preserve the consistency of image content in different domains by using two GANs and a cycle consistency loss to achieve this characteristic.

3.1.1 CycleGAN

During the training of CycleGAN, we use images from two different image domains for training, such as night and day. In our study, we consider nighttime images as domain X and daytime images as domain Y. Our goal is to train two generator models that convert domain X images to domain Y images and domain Y images to domain X images respectively. The generator of CycleGAN mainly uses the Auto-encoder architecture. Generator A converts domain X images into domain Y images, and generator B converts domain Y images into domain X images. Furthermore, CycleGAN combines two discriminator models to evaluate the authenticity of generated images. Discriminator A is used to judge whether the input image is a real image from domain X or a fake image generated by generator B, while discriminator B is used to judge whether the input image is a real image from domain Y or a fake image generated by generator A. The ground truth label of the real image is 1, and the output of the discriminator for a real image needs to be as close to 1 as possible. The ground truth label of the fake image is 0, and the output of the discriminator for the fake image needs to be as close to 0 as possible. The generator needs to generate fake images that make the output of the discriminator as close to 1 as possible.

The generator and the discriminator compete against each other through adversarial training. The generator is driven by the adversarial loss function to produce realistic images, preventing the discriminator from being able to differentiate generated images from actual ones. CycleGAN uses a cycle consistency loss to maintain the structure of the image. The cycle consistency loss makes sure consistency in the bidirectional transformation process. After the image enters a generator, it will generate a fake image of the other domain, and

then the fake image will enter another generator to output a reconstructed image. The cycle consistency loss is used to determine the similarity of the reconstructed image to the original image. To drive the training process, CycleGAN trains the model using the two loss functions mentioned above. The generator and discriminator models will advance one another through ongoing iterative training, making the generated image closer to the target domain while maintaining the characteristics and structure of the original image.

3.1.2 CycleGAN for Night and Day

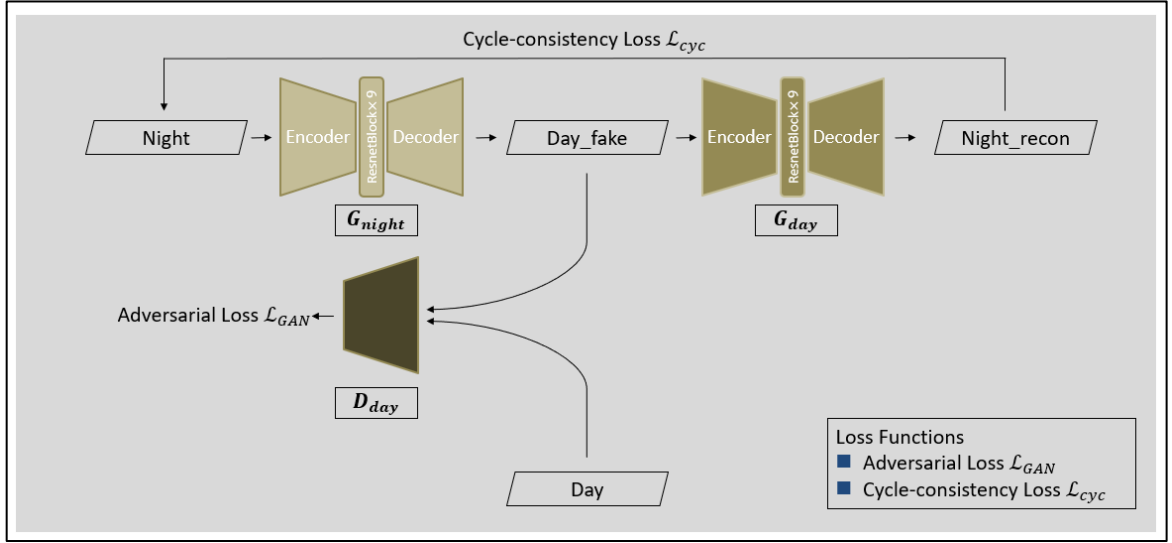


Figure 2. The architecture of the proposed domain adaption model training.

The lighting conditions and ambiance vary greatly between day and night. In our strategy, we want to train a CycleGAN model that can convert day to night and vice versa. About 10,000 day and night images from the CULane lane dataset are used to train the CycleGAN model. The model to be trained contains two generators (G_{night} and G_{day}) and two discriminators (D_{night} and D_{day}). The G_{night} generator transforms night images into day and the G_{day} generator transforms day images into the night. D_{night} and D_{day} discriminators are used to identify real and fake images in day and night domains, respectively. Equation (1) represents the adversarial loss from the night domain (X) to the day domain (Y):

$$L_{GAN}(G_{night}, D_{day}, X, Y) = E_{y \sim p_{data}(y)}[\log D_{day}(y)] + E_{x \sim p_{data}(x)}[\log(1 - D_{day}(G_{night}(x)))] \quad (1)$$

where D_{day} tries to distinguish if the image is a real daytime image or a synthetic daytime image, and G_{night} attempts to generate a synthetic image of daytime that closely resembles the original image. Similarly, another adversarial loss is needed to focus on the translation from the day domain (Y) to the night domain (X), namely $L_{GAN}(G_{day}, D_{night}, Y, X)$. In CycleGAN, an adversarial loss is used to help the generated image better match the distribution of the target domain. Additionally, a cycle consistency loss is introduced to ensure that the image transformation is reversible; this allows for the use of the generated image to reconstruct the original image. The idea is to guarantee the integrity and consistency of the image during the conversion process. We define the cycle consistency loss as Equation (2):

$$\begin{aligned} L_{cyc}(G_{night}, G_{day}) &= E_{x \sim p_{data}(x)}[\|G_{day}(G_{night}(x)) - x\|_1] \\ &+ E_{y \sim p_{data}(y)}[\|G_{night}(G_{day}(y)) - y\|_1] \end{aligned} \quad (2)$$

where G_{night} tries to convert night images into day images, and G_{day} tries to convert day images into night images. Our overall objective function is shown in Equation (3).

$$\begin{aligned} L_{GAN}(G_{night}, G_{day}, D_{night}, D_{day}) &= L_{GAN}(G_{night}, D_{day}, X, Y) + L_{GAN}(G_{day}, D_{night}, Y, X) \\ &+ L_{cyc}(G_{night}, G_{day}) \end{aligned} \quad (3)$$

This function contains two adversarial losses and one cycle consistency loss and is used to train two generators and two discriminators.

3.2 Lane Detection Model Training

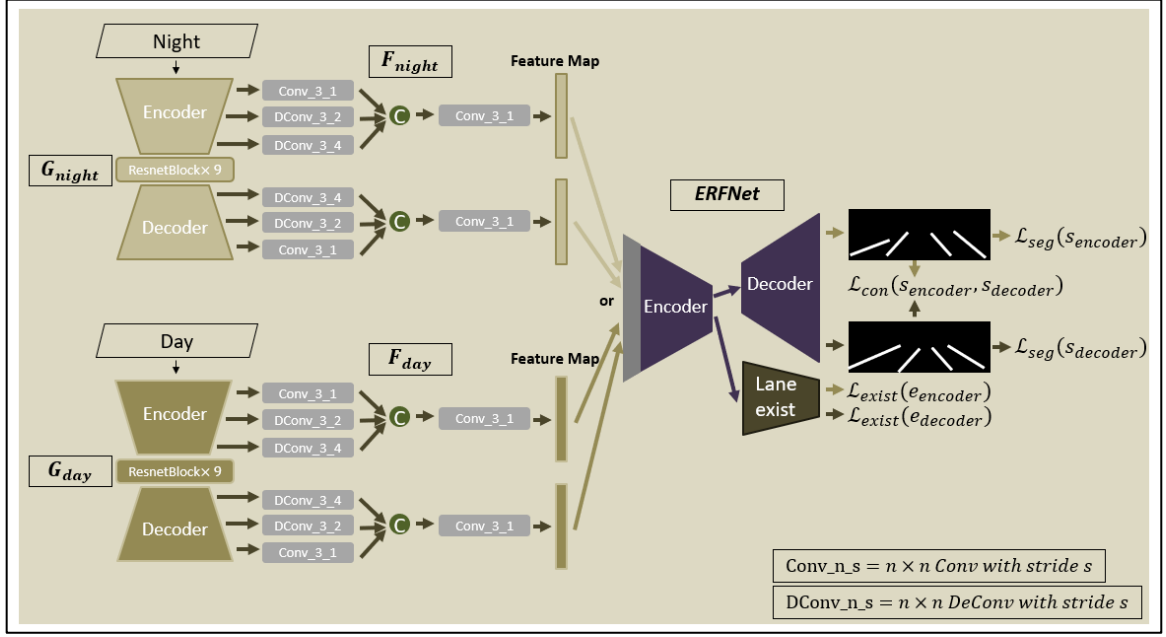


Figure 3. The architecture of the proposed lane detection model training.

Based on the method of the first stage, we have trained a domain adaptation model based on CycleGAN to convert night and day images. The overall architecture of our proposed lane detection model is shown in Figure 3, which consists of a domain adaptation module, feature fusion module, and image segmentation module. Among them, the domain adaption module is used to input night or day images and outputs the features representing night and day. The image segmentation module receives the fused upper and lower features as input after the feature fusion module has extracted multi-scale features. The image segmentation module is based on the existing advanced image segmentation model and outputs the final lane prediction.

3.2.1 Domain Adaption Module

In our proposed method, the domain adaptation module can be used to generate multi-scale feature maps representing night and day so that our model can be trained to feature maps of various scales and multiple scenes, which will help enhance the accuracy of the model and robustness. We use the generators of already trained domain adaptation models

as our domain adaptation modules, namely G_{night} and G_{day} in Figure 2. We will extract feature maps of three scales each from the Encoder and Decoder of the domain adaptation module. In the domain adaption module, whether it is Encoder or Decoder, we regard a convolutional layer, an instance normalization layer, and a ReLU activation function as a generator block. Let B^i be a general function representing a generator block, and the generator block will output the feature map of the next layer. Suppose the dimension of the i -th layer feature map is $[w_i \times h_i \times c_i]$, w is the width of the feature map, h is the height of the feature map, and c is the number of channels output by the generator block. As Equation (4) shows, $B_{encoder}^i$ is the generator block representing the encoder in the generator. The feature map f_i generated by the Encoder is shown in Equation (4):

$$\begin{aligned} f_i &= B_{encoder}^i(f_{i-1}) \\ \text{Where, } f_i &= \mathbb{R}^{w_i \times h_i \times c_i} \\ \text{and, } w_i &= W/2^{i-1}; h_i = H/2^{i-1}; c_i = 2^{5+i}; i = 1, 2, 3 \end{aligned} \quad (4)$$

where f_0 is the image input into the model, and the size is $W \times H$. W and H are the width and height of the input image, respectively. As shown in Figure 2, in the CycleGAN generator, there is a module called Resnetblock between the Encoder and the Decoder. There are 9 residual blocks in a Resnetblock, and the dimensions of each block's input and output are the same. In our definition, the input of Resnetblock is f_3 . We define the entire Resnetblock as B_{resnet} , and the feature map f_i generated by Resnetblock is shown in Equation (5):

$$\begin{aligned} f_i &= B_{resnet}(f_{i-1}) \\ \text{Where, } f_i &= \mathbb{R}^{w_i \times h_i \times c_i} \\ \text{and, } w_i &= w_{i-1}; h_i = h_{i-1}; c_i = c_{i-1}; i = 4 \end{aligned} \quad (5)$$

The feature map output by Resnetblock is f_4 , which has the same dimension size as f_3 , and will be used as the input of the CycleGAN decoder. $B_{decoder}^i$ is the generator block representing the decoder in the generator. The feature map f_i generated by the Decoder is shown in Equation (6).

$$\begin{aligned}
f_i &= B_{decoder}^i(f_{i-1}) \\
\text{Where, } f_i &= \mathbb{R}^{w_i \times h_i \times c_i} \\
\text{and, } w_i &= W/2^{6-i}; h_i = H/2^{6-i}; c_i = 2^{12-i}; i = 5, 6
\end{aligned} \tag{6}$$

3.2.2 Feature Fusion Module

We have extracted feature maps of various scales from the Encoder and Decoder of the domain adaptation module. As shown in F_{night} and F_{day} in Figure 2, The primary purpose of the feature fusion module is to connect the domain adaptation module and the image segmentation module in series. This module fuses multi-scale feature maps and outputs feature maps representing night and day. In order to integrate multi-scale features, feature maps of different scales will enter a convolution kernel to adjust the feature maps of different scales to the same size, and adjust the number of channels to the same size to average the influence of each feature map. After validation in our internal experiments, we choose to output feature maps with the same dimensions as the image. After the convolution kernel, The three feature maps are connected, and a convolution kernel is used to change the number of channels, and output the final feature map to the image segmentation module. Since we use feature maps that include Encoder and Decoder, our feature fusion module will eventually output two feature maps to represent different scenes. The following is the operation process of the feature fusion module. The feature map g_i is shown in Equation (7):

$$\begin{aligned}
g_i &= B_{CBL_{n_s}}(f_i) & i = 1, 6; n = 3; s = 1 \\
g_i &= B_{DCBL_{n_s}}(f_i) & i = 2, 5; n = 3; s = 2 \\
g_i &= B_{DCBL_{n_s}}(f_i) & i = 3, 4; n = 3; s = 4
\end{aligned} \tag{7}$$

where f_i is the feature map extracted by the domain adaptation model. $B_{CBL_{n_s}}$ is represented as a convolutional block containing a convolutional layer, batch normalization layer, and a leakyReLU activation function. $B_{DCBL_{n_s}}$ is represented as a convolutional block containing a convolutional layer, batch normalization layer, and a leakyReLU activation function. The convolution kernel size is represented by n , and s is the stride.

g_i is the feature map after the first convolution layer. The feature map f_{concat_en} and f_{concat_de} is shown in Equation (8):

$$\begin{aligned} f_{concat_en} &= F_{concat}(g_1, g_2, g_3) \\ f_{concat_de} &= F_{concat}(g_4, g_5, g_6) \end{aligned} \quad (8)$$

where F_{concat} denotes the concatenate operation. f_{concat_en} and f_{concat_de} are the feature maps connected by the three feature maps output by the Encoder and Decoder, respectively.

The feature map $f_{encoder}$ and $f_{decoder}$ is shown in Equation (9):

$$\begin{aligned} f_{encoder} &= B_{CBL_n_s}(f_{concat_en}) \\ f_{decoder} &= B_{CBL_n_s}(f_{concat_de}) \end{aligned} \quad n = 3; s = 1 \quad (9)$$

where f_{concat_en} and f_{concat_de} need to go through a convolution block respectively. The $f_{encoder}$ and $f_{decoder}$ are the final output of the feature fusion module and are also the input of the image segmentation module.

3.2.3 Image Segmentation Module

In the previous chapter, we got the feature map combining Encoder and Decoder. Next, we use the image segmentation module to input feature maps representing night and day to detect the position and shape of the lane. The image segmentation module can choose advanced image segmentation network models such as ERFNet [25], DLA [34], and DDRNet [8]. An image segmentation model is a deep learning technique that aims to segment an input image into different regions or groups of pixels and assign a corresponding label to each region. This technique enables the model to identify and distinguish between different objects, structures, or features in the imagery, enabling precise area localization and classification. Image segmentation models usually use deep learning architectures such as convolutional neural networks (CNNs) to learn to extract features from a large amount of labeled training data, and then achieve accurate segmentation and classification tasks for new images. In lane detection, the image segmentation model can help us effectively identify different lane lines and backgrounds, thus providing key information support for lane

detection methods. We use ERFNet as an example to illustrate the changes in our method and model. ERFNet is a deep learning model for semantic segmentation tasks. The core idea of ERFNet is to use factorized convolutions and residual connections to achieve efficient and accurate models. Factorized convolution splits a large convolution operation into multiple smaller convolution operations to reduce the computational load of the model. The Non-bottleneck-1D they proposed in the Encoder section is a module that combines factorized convolution and residual connection and expands the field of view of the convolution layer by adding dilated convolution.

The traditional lane detection method directly uses RGB images as the input of the image segmentation model. In our method, we input feature maps with 16 channels into the model. To do this, we change the number of input channels from 3 to 16 in the first layer of the ERFNet model. In our approach, since we only need to segment out different lanes and backgrounds, ERFNet is by definition an instance segmentation model. The CULane dataset we use will only have up to 4 lane lines at the same time, so we set the number of channels output by the last layer of the model to 5, representing 4 lane lines and the background respectively. According to the method of ENet-SAD [9], in order to accelerate the convergence speed of the network during training and to output the lanes with confidence values higher than the threshold during testing. We add a lane branch network parallel to the Decoder in the ERFNet model. The branch network includes convolutional layers and fully connected layers, and finally outputs four floating-point numbers from 0 to 1, corresponding to the confidence value of each lane.

The original ERFNet model uses two loss functions as feedback for network training during lane detection network training. The first is the instance segmentation map output by the decoder, which represents the probability of whether each pixel is a lane or background. The confidence of the existence of the corresponding lane is output by the second branch

lane existence branch, and the loss function is shown in Equation (10):

$$L_{lane} = \lambda_1 L_{seg}(s, \hat{s}) + \lambda_2 L_{exist}(e, \hat{e}) \quad (10)$$

where L_{seg} is the negative log-likelihood loss function for instance segmentation, and L_{exist} is the lane existence binary cross-entropy loss function. s is the instance segmentation map output by the decoder, $\hat{s}$ is the ground truth of the instance segmentation map. e is the confidence of the existence of the lane, the value is between 0 and 1, and $\hat{e}$ is the ground truth of the lane existence, if the lane exists, it is represented by 1, and if the lane does not exist, it will be represented by 0. We set weights $\lambda_1 = 1$, $\lambda_2 = 0.1$.

3.2.4 Consistent Loss

Our model will eventually output model predictions from the feature maps of CycleGAN Encoder and Decoder. Although these two output predictions are the results of the same image input, there will still be differences in the output of the predicted lane pixel probability map. Therefore, our proposed consistency loss function constrains the variance of model outputs. The two domain adaptation modules we have trained can convert night images into day and day images into night respectively. This means that for the night-to-day domain adaption module, the feature map extracted from the Encoder is closer to the features of the night, and the feature map extracted from the Decoder is closer to the features of the day. For the domain adaption module that changes from day to night, it will extract feature maps close to day and night, respectively. We expect that by restricting the input of the two feature maps to the lane pixel probability map output by the image segmentation module, the image segmentation module can output nearly consistent results under different light conditions, which can improve the performance and robustness of the neural network model. Ability to detect lane markings that are difficult to recognize based on different lighting conditions.

To this end, we propose an algorithm for the consistency loss function. We use the

common L1 loss function. L1 loss is the mean absolute error (MAE) between each element in the measured input. In our approach, it is used to compute the per-pixel difference in the image. The loss function is shown in Equation (11):

$$L_{con} = L_1(s_{encoder}, s_{decoder}) \quad (11)$$

where $s_{encoder}$ and $s_{decoder}$ respectively represent the instance segmentation map predicted by the feature map extracted by Encoder and Decoder.

3.2.5 Training Process

This section explains how to combine domain adaptation, feature fusion, and Image segmentation modules for lane detection training. As shown in Figure 2, we will divide the training images into nighttime and daytime images. Nighttime images will be input into the CycleGAN generator that turns night into day, namely G_{night} , and daytime images will be input into the CycleGAN generator that turns day into night, namely G_{day} . The multi-scale feature maps output by G_{night} and G_{day} will be input to the respective feature fusion modules, namely F_{night} and F_{day} . The $f_{encoder}$ and $f_{decoder}$ output via F_{night} or F_{day} will be input to the lane detection module at the same time. The lane detection module will eventually output the instance segmentation maps $s_{encoder}$ and $s_{decoder}$ output by the decoder at the same time, as well as the confidence $e_{encoder}$ and $e_{decoder}$ of the lane existence. Equation (12) is the total loss of the lane detection model.

$$\begin{aligned} L_{total} = & \lambda_1 \left(L_{seg}(s_{encoder}, \hat{s}) + L_{seg}(s_{decoder}, \hat{s}) \right) \\ & + \lambda_2 \left(L_{exist}(e_{encoder}, \hat{e}) + L_{exist}(e_{decoder}, \hat{e}) \right) \\ & + \lambda_3 L_{con}(s_{encoder}, s_{decoder}) \end{aligned} \quad (12)$$

We set weights $\lambda_1 = 1$, $\lambda_2 = 0.1$, $\lambda_3 = 1$.

Next, we will introduce our training process. Since our training data will be divided into night and day images. During the same epoch, we will first train the night images and then train the day images. The night images and day images have their data loaders to shuffle

their respective datasets. Through our internal experiments, updating the model parameters of G_{night} and G_{day} does not help to improve the performance of the overall model, so when training night images, the Optimizer will only update the model parameters of F_{night} and ERFNet. When training daytime images, the Optimizer will only update the model parameters of F_{day} and ERFNet.

Chapter 4 Experimental Evaluation

In this chapter, detailed experimental data and settings are provided, as well as the results and analysis of the conducted experiments. In Section 4.1, we describe the dataset used, and the preprocessing steps applied to the images, and provide evaluation metrics for analyzing model performance. We also detail the hardware and software setup we used to train and test the models. In Section 4.2, we discuss preliminary experiments performed to explore potential ways to improve our study. We analyzed the results of these experiments and their testing value. In Section 4.3, we present the in-house experiments performed to identify the best of our proposed methods. Furthermore, in Section 4.4, we conduct external experiments on the CULane dataset, focusing on different and challenging scenarios.

4.1 Experimental Data and Settings

The proposed approach using a feature fusion architecture combined with a domain adaption model and an image segmentation model applies to any semantic segmentation-based lane detection model. We carried out experiments on the CULane [23] dataset to evaluate the performance of the proposed method. This dataset is among the most comprehensive lane detection datasets available, comprising 88,880 images for training and 34,680 images for testing. The test images contain several challenging driving scenarios, including nighttime and other challenging lighting conditions. The original size of the images in the CULane dataset is 1640×590 . We refer to the experimental settings of the SIM-CycleGAN [20] study, crop the image by 240 pixels from the top of the image, and resize the image to 976×208 . During the training and testing using the DLA model, based on the downsampling requirements of the DLA model, the input image size is 992×224 .

In order to evaluate our method objectively, we follow the literature SCNN [24] to evaluate the evaluation metrics of the CULane dataset. After our model predicts the instance

segmentation map of the lane, we use the procedure of SCNN [24] to convert the instance segmentation map to point coordinates in the same format as the ground truth label. Treat both the predicted and ground-truth point coordinates as lane lines with a width of 30 pixels, and calculate the intersection union (IoU) between the predicted and ground-truth labels. IOU greater than 0.5 is regarded as true positive (TP). Misjudged lanes are regarded as false positive (FP), and missed lanes are regarded as false negative (FN). We calculate Precision, Recall, and F1 score based on the above indicators.

IoU represents the overlap ratio of the predicted lane to the ground truth of the lane, and according to SCNN [24], we consider the lane as a plane with a width of 30 pixels to calculate the overlap ratio. If the IoU is greater than 0.5, it is regarded as True positive (TP). The higher the IoU, the more accurate the predicted position of the lane. The formula is shown in Equation (13):

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (13)$$

Precision refers to the proportion of the model that correctly predicts positive samples in the predicted positive samples. In other words, precision quantifies how many of the samples predicted as positive are actually true positives. The formula is shown in Equation (14):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (14)$$

Recall refers to the proportion of correctly predicted positive samples in the total true positive samples. The formula is shown in Equation (15):

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (15)$$

F1 score is the harmonic mean of precision and recall, considering both precision and recall combined metrics. The F1 score balances the trade-off between precision and recall.

The formula is shown in Equation (16):

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (16)$$

The training and testing process was executed on two NVIDIA 3090 graphic processing units. The CUDA version was 11.2, the Python version was 3.8.13, and the Pytorch version was 1.12.0. For the training of the CycleGAN model, we used approximately 10,000 day and night images from the CULane dataset. The Adam optimizer was implemented with an initial learning rate of 0.0002, and training was performed for 200 epochs. it took approximately 3-4 days to complete. To train the lane detection model, we utilized the SGD optimizer with an initial learning rate of 0.01 and performed training for 24 epochs. The training time is about 1-2 days.

4.2 Preliminary Experiments

In our research, we tried some possible intuitive ways to improve our research, and we learned that the results of some experiments may not meet the expected performance level. Nevertheless, we believe that these experiments still have important testing value and deserve to be described and discussed in this study.

4.2.1 Data Augmentation Method

Our main method is to train the domain adaptation model first, and then use the trained domain adaptation model to join the lane detection model for training. In this experiment, we intuitively use the trained domain adaptation model to generate synthetic day and synthetic night images and use the generated images to join the training of the lane detection model. The experimental results in Table 4, show that the data augmentation method does not improve the recognition ability of the model in our method. In the experiment, we also tried to use different weights to determine the extent to which the augmented data affects the model, but the performance of the model still could not be improved. We think that the

possible reason is that after training with augmented data, the model may not be able to focus on the feature extraction of the original image. Therefore, in subsequent experiments, we will only use the original training data set for training.

Table 4. Data Augmentation method comparison.

Models	Data Augmentation	Augmentation Data Loss Weight	F1-score (%)	
			Total	Night
ERFNet			73.6	68.5
	✓	1	68.4	59.7
	✓	0.5	68.6	57.7
	✓	0.1	69.4	59.0

4.2.2 Domain Adaption Model Variants

For the training of the domain adaptation model, theoretically, a better domain adaptation model can better extract features during the second stage of lane detection model training, which is helpful for the training of the overall model. Therefore, we tried several methods to improve our domain adaptation model. For the first approach, we refer to CIGAN [23] for adding noise to the domain adaptation model. We add random noise based on the normal distribution to the output of each layer of the Decoder in the day-to-night Generator and then multiply it by a learnable weight to make the feature distribution more diverse and synthesize a more realistic nighttime image. In the second method, we try to replace the CycleGAN Generator originally based on the ResNet [36] network architecture with other backbone networks. We believe that using the new backbone network may have the opportunity to make the model training of CycleGAN more efficient and achieve better results. Therefore, we try to transform the backbone network of MobileNetv2 [27] and EfficientNet [31] into the network architecture of Auto-encoder and become the network architecture of CycleGAN Generator.

According to the experimental results in Table 5, these methods have not achieved the expected results of the model. We think that the method of adding noise may be reasonable in the method of training the domain adaptation model, but it will not help the extraction of features when training the lane detection model. In addition, since the noise is added to the entire image, it may make the lane lines less obvious. For the method of replacing the backbone network, we think that the possible reason is that this efficient backbone network may not be suitable for training the domain adaptation model. However, since the architecture of the network was created by ourselves, we did quite limited tests and maybe this approach still has potential. But based on the experimental results, we still use the original CycleGAN to implement our method and conduct follow-up experiments.

Table 5. Domain adaption model variants comparison.

Models	Noise	Backbone	F1-score (%)	
			Total	Night
CycleGAN		ResNet	73.6	68.5
	✓	ResNet	72.8	66.1
		MobileNetV2	71.2	60.0
		EfficientNet	72.6	65.2

4.2.3 Image pre-processing

In our research, we also try to explore not using GAN for image style conversion, but using image pre-processing technology to solve the difficult problem of night scene detection. Specifically, we tried two methods, Histogram Equalization (HE) and Contrast Limited Adaptive Histogram Equalization (CLAHE), trying to brighten the night image to a state close to the daytime. We applied the above image preprocessing method to the night image in the test dataset. We transferred the image to the Hue, Saturation, Value (HSV) coordinate system, and adjusted the Value representing the brightness through the above two methods.

Next, we use the pre-processed images instead of the original night images as part of the test dataset and test on the original ERFNet. This step is to evaluate whether simply changing the image brightness can improve the lane detection ability of the night image. However, our experimental results in Table 6 show that only adjusting the brightness of the image cannot effectively improve the lane detection ability of the night image. Although the brightness of the image may thus be enhanced, there are still difficulties in lane detection due to problems such as uneven light brightness and image noise. This result highlights the complexity of style transfer and the limitation of a single brightness adjustment for solving the lane detection problem in night images.

Table 6. Image pre-processing comparison.

Models	Version	F1-score (%)	
		Total	Night
ERFNet		73.1	67.1
	HE	71.3	57.7
	CLAHE	71.0	55.8

4.3 Internal Experiments

To confirm that our model can achieve the best performance, we performed several internal experiments to try to figure out which method performed better. Our internal experiments include whether to train the domain adaptation model in conjunction with the lane detection model, whether to update only some parameters when training the lane detection model, and the size of the feature map output by the feature fusion module.

4.3.1 Domain adaption model training strategy

Table 7. Domain adaption model training strategy comparison.

Models	Image Segmentation module	Feature Fusion Module	Freeze (Image Segmentation module)	F1-score (%)	
				Total	Night
CycleGAN				73.6 (+0.5)	68.5 (+1.4)
	✓			73.2 (+0.1)	67.8 (+0.7)
	✓		✓	73.2 (+0.1)	67.8 (+0.7)
	✓	✓		72.8 (-0.3)	66.6 (-0.5)
	✓	✓	✓	72.7 (-0.4)	66.8 (-0.3)

In order to train the best domain adaptation model, we not only try to use the original CycleGAN to train the domain conversion model that converts between day and night but also try to connect the CycleGAN model to the lane detection model. We expect to use the lane detection loss to constrain CycleGAN to generate synthetic images while maintaining the features and shapes of lanes as much as possible. Regarding the way of concatenating the lane detection model, we also use the ERFNet model as the lane detection model, and we have also tried two methods. The first is to directly enter the synthetic image generated by CycleGAN into the lane detection model, and the second is to combine CycleGAN with the feature fusion module to output the feature map into the lane detection model. We also tested the lane detection models of the above two methods as frozen weights. We input the weights of the trained model into the domain adaptation module of our proposed method and test which method has the best lane detection result.

The experimental results in Table 7 show that the model trained directly using the original CycleGAN has the best results. We speculate that this may mean that even if the lane detection model is not connected in series to restrict the training of the model, the

original CycleGAN can also preserve the characteristics of the lane line well while transferring the domain. On the contrary, the lane detection loss in the method of concatenating lane detection models may not allow the model to focus on domain translation.

Therefore, in the following experiments, we still directly use the original CycleGAN as the domain adaptation model.

4.3.2 Optimizer strategy

Table 8. Optimizer strategy comparison.

Models	Optimizer			Consistent Loss	F1-score (%)	
	Domain Adaption Module	Feature Fusion Module	Image Segmentation module		Total	Night
ERFNet	✓	✓	✓		72.1 (-1.0)	67.1 (+0.0)
				✓	73.3 (+0.2)	68.6 (+1.5)
		✓	✓		73.6 (+0.5)	68.5 (+1.4)
				✓	74.3 (+1.2)	69.1 (+2.0)
			✓		70.5 (-2.6)	62.3 (-4.8)
				✓	71.1 (-2.0)	62.4 (-4.7)

In the lane detection model training phase, since our model includes a domain adaptation module, feature fusion module, and image segmentation module, whether to update all the modules during the training phase or to freeze the parameters of some modules is a question worth discussing. We tried three ways to update the modules. The first method is to update all modules, the second method is to update only the image segmentation module, and the third method is to update only the feature fusion module and image segmentation module. The experimental results in Table 8 show that it is the best method to only update the feature fusion module and the image segmentation module and freeze the parameters of the domain adaptation module. Freezing the parameters of the domain adaptation module

allows the module to focus on extracting the night and day features of the image without being affected by the loss of lane detection. Freezing part of the parameters can reduce GPU memory usage and help the model converge faster, which may be the reason why this method performs slightly better than other methods.

4.3.3 Output Size of the Feature Map

During the training phase of the lane detection model, the feature fusion module will extract feature maps of different sizes generated by the domain adaptation module, and finally output the feature maps to the image segmentation module. As a method proposed by ourselves, the feature fusion module needs more tests to adjust the model architecture. In this experiment, we try to adjust the architecture of the module to output feature maps of different sizes, the first size is the same as the original input image size, and the second size is half the size of the input image. The experimental results in Table 9 show that feature maps with the same size as the input image can perform better. Although the feature map with the same input image size requires more computation, it can also retain more features to the image segmentation module, making the overall model perform better.

Table 9. Output size of the feature map comparison.

Models	Feature Map Size	Consistent Loss	F1-score (%)	
			Total	Night
ERFNet	Half Size (488 × 104)		73.5 (+0.4)	67.9 (+0.8)
		✓	73.8 (+0.7)	67.2 (+0.1)
	Same Size (976 × 208)		73.6 (+0.5)	68.5 (+1.4)
		✓	74.3 (+1.2)	69.1 (+2.0)

4.3.4 Qualitative analysis of Grad CAM

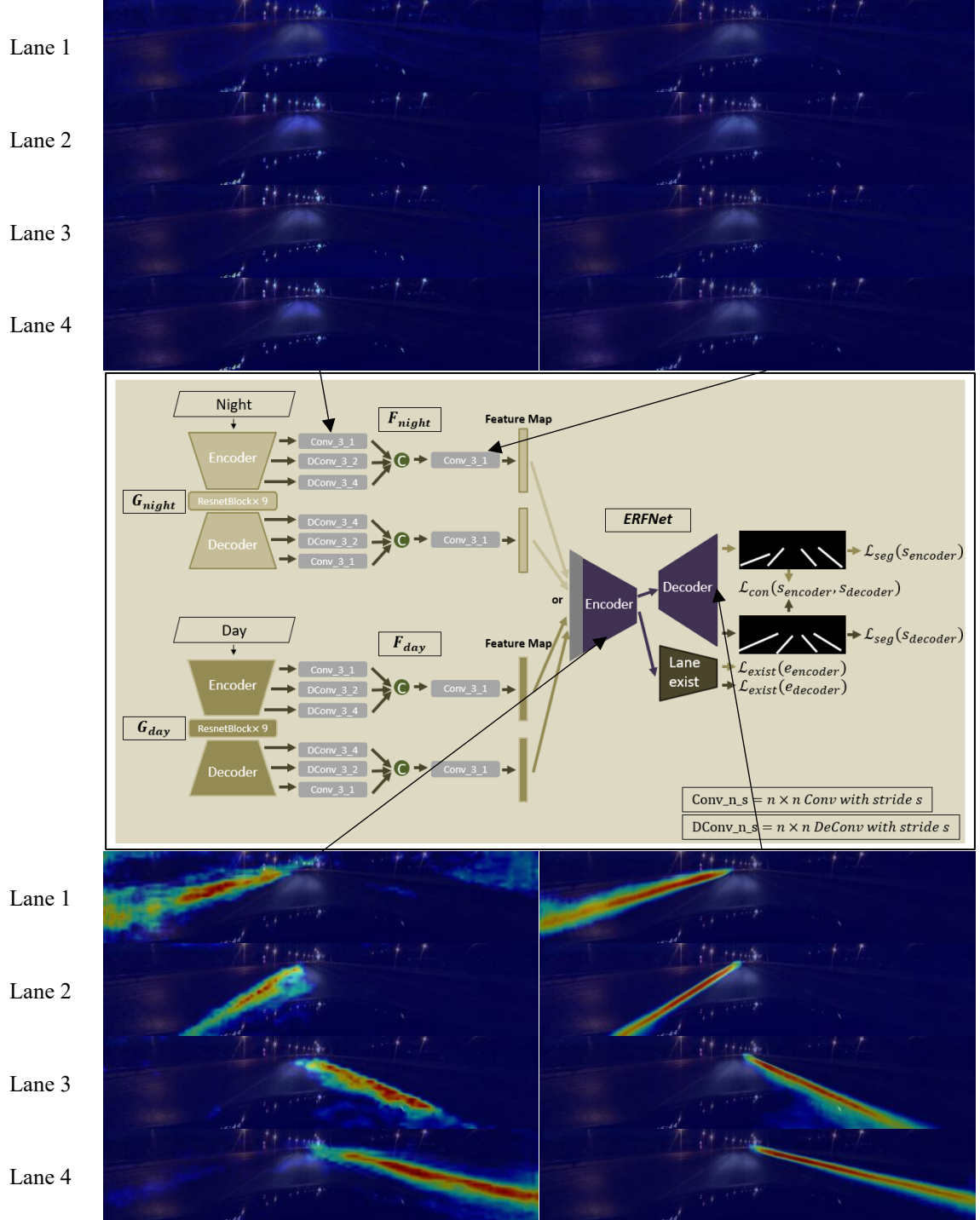


Figure 4. Qualitative analysis result of Grad CAM.

Figure 4 shows that we used the Gradient-weighted Class Activation Mapping (Grad-CAM) technique for qualitative analysis to deeply study whether the lane detection model we trained pays attention to lane features at various stages of different modules. First, we

analyzed the outputs of the four lanes at two different locations of the feature fusion module. After analysis, in the two layers of the feature fusion module, the distribution of the heat map is not concentrated in the lane area. This means that the feature fusion module seems to pay little attention to lane features at these levels. We also output heat maps for analysis in the encoder and decoder of the image segmentation module. In the part of the encoder and decoder, the distribution of heat maps is indeed concentrated in the lane area. This shows that in these layers, the model does pay attention to and learn lane-related features. As for why only the encoder and decoder parts have an obvious focus on the lane, the possible reason is that in the encoder part, the size of the feature map has been compressed to a very small size, making it impossible for the shallower network layer to focus on extracting the features of the lane. In general, from the results of Grad-CAM qualitative analysis, our lane detection model may not pay attention to lane features in the feature fusion module, but it has lanes in the encoder and decoder parts of the image segmentation module feature attention.

4.4 External Experiments

In order to confirm that our method can effectively improve the performance of the original lane detection model. Our external experiments consisted of using our method with three different models, testing the performance on the CULane challenging scene and additional challenging scenes of our own choice.

4.4.1 CULane Total & Night

Table 10. CULane total and night comparison.

Models	Version	F1 score (%) (Precision/Recall) (%)	
		Total	Night
ERFNet	Original	73.1 (73.4/72.9)	67.1 (67.5/66.8)
	Ours	73.6 (75.4/72.0)	68.5 (74.2/63.6)
	Ours(Con:L1)	74.3 (76.1/72.6)	69.1 (73.3/65.4)
DLA	Original	71.5 (71.7/71.2)	64.4 (65.4/63.4)
	Ours	73.1 (73.7/72.5)	66.6 (68.2/65.1)
	Ours(Con:L1)	73.9 (76.1/71.9)	68.1 (72.7/64.0)
DDRNet	Original	68.9 (71.8/66.3)	59.9 (67.0/54.2)
	Ours	69.4 (76.3/63.7)	61.7 (73.1/53.3)
	Ours(Con:L1)	69.6 (76.1/64.2)	62.2 (74.1/53.6)

Table 10 shows the F1 score, precision, and recall of the proposed method on the CULane dataset in different models. The table lists the performance difference between our method and the original model on total and night scenes. We choose three models to use our

method, which is to connect the domain adaptation module and feature fusion module to these models. In addition, the final version of our method also adds a consistency loss to assist the training of the model. Experiments show that our proposed method can effectively improve the performance of the model. The F1 score of the original ERFNet model in the CULane test data set is 73.1, and the F1 score in the night scene is 67.1. In our method, the overall F1 score of the ERFNet-based model is 73.6, and the F1 score of the night scene is 68.5. After adding Consistency Loss, the F1 score increased to 74.3, and the night scene increased to 69.5. We also test our method with DLA and DDRNet models, and their overall F1 score can be improved by 0.5 to 2.4, and the F1 score in night scenes can be improved by 2.3 to 3.7. After further analysis of Precision and recall, it can be found that although the recall is sometimes slightly lower than the original model, the performance of Precision can be greatly improved. Overall, This proves that our method is applicable to different image segmentation models and can effectively improve the performance of the model.

4.4.2 CULane Challenge Scene

In the CULane test data set, in addition to the night scene, there are other challenging scenes, including Crowded, Dazzle, Shadow, and Arrow. Crowded represents a scene with many vehicles. Because too many vehicles will increase the probability of lane lines being obscured, it will increase the difficulty of detecting lanes. Dazzle refers to scenes with glare, which usually appear in the early morning or evening. Excessive strong light will not only reduce the quality of the entire image but also blur some lane lines, which will affect lane detection performance. Shadow refers to lane lines under bridges or other shadows. Shadows will reduce the contrast of the image, and the edges of lane lines may become less clear. Arrow refers to objects similar to lane lines such as pointing lines in the lane, which may cause the model to misjudge. We also use the ERFNet model to verify that our method can effectively improve the recognition ability of the model in these challenging scenarios. The

experiments in Table 11 show that our method can effectively improve the recognition ability of the model in these scenes, and the F1 score has increased by about 0.1 to 1.2%. After adding Consistency Loss, the F1 score can be increased by 0.5 to 3.5% in total.

Table 11. CULane challenge scene comparison.

Models	Version	F1 score (%) (Precision/Recall) (%)			
		Crowded	Dazzle	Shadow	Arrow
ERFNet	Original	71.6 (72.4/70.8)	65.6 (66.3/64.9)	71.4 (71.7/71.3)	87.1 (88.7/85.5)
	Ours	71.7 (72.9/70.6)	66.0 (68.9/63.4)	72.6 (73.6/71.6)	87.6 (89.6/85.6)
	Ours(Con:L1)	72.1 (73.7/70.6)	66.3 (69.6/63.2)	74.9 (76.5/73.4)	88.2 (90.0/86.5)

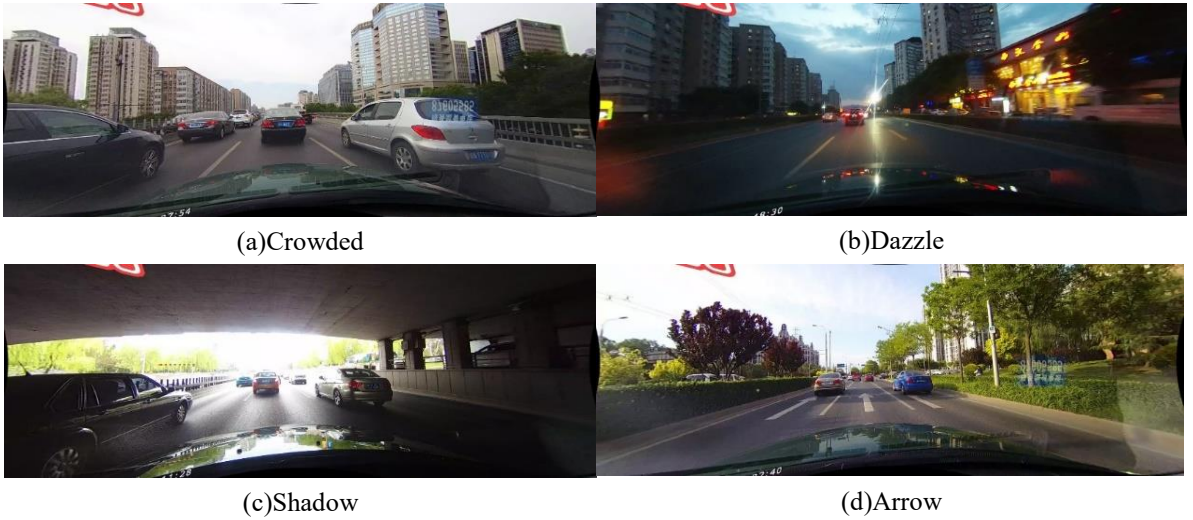


Figure 5. The example of challenge scenarios from the CULane test dataset.

4.4.3 Other CULane Challenge Scene

In order to verify that our method can have a more stable recognition ability in other challenging scenes. We self-selected two special scenarios to test our method on the CULane test set. The first scene is an indoor scene, including underground parking lots and tunnels, with a total of 102 images. Indoor scenes usually lack natural light sources, which is different

from daytime scenes, which have a large proportion of the training data. The strong light of some artificial lighting may also affect the detection of lane markings. The second scene is the wet ground scene, and there are 240 images in total. The wet ground will change the reflection characteristics of the road surface, which may cause the edges and color features of lane markings to become unclear, thus affecting the accuracy of lane marking detection. We also use the ERFNet model to verify that our method can effectively improve the relatively stable identification ability of the model in these special scenarios. The experiments in Table 11 show that our method can effectively improve the recognition ability of the model in these scenes, and the F1 score of indoor scenes has increased by about 1.5 to 2.1%. The F1 score of the wet ground scene can be increased by about 5.2 to 8.2%.

Table 12. CULane other challenge scene comparison.

Models	Version	F1 score (%) (Precision/Recall) (%)	
		Indoor	Wet Ground
ERFNet	Original	48.5 (56.3/42.6)	54.4 (50.2/59.3)
	Ours	50.0 (67.5/39.7)	59.6 (63.4/56.2)
	Ours(Con:L1)	50.6 (70.7/39.4)	62.6 (64.7/60.7)



(a)Indoor

(b)Wet Ground

Figure 6. The example of other challenge scenarios from the CULane test dataset.

4.4.4 Results Visualization

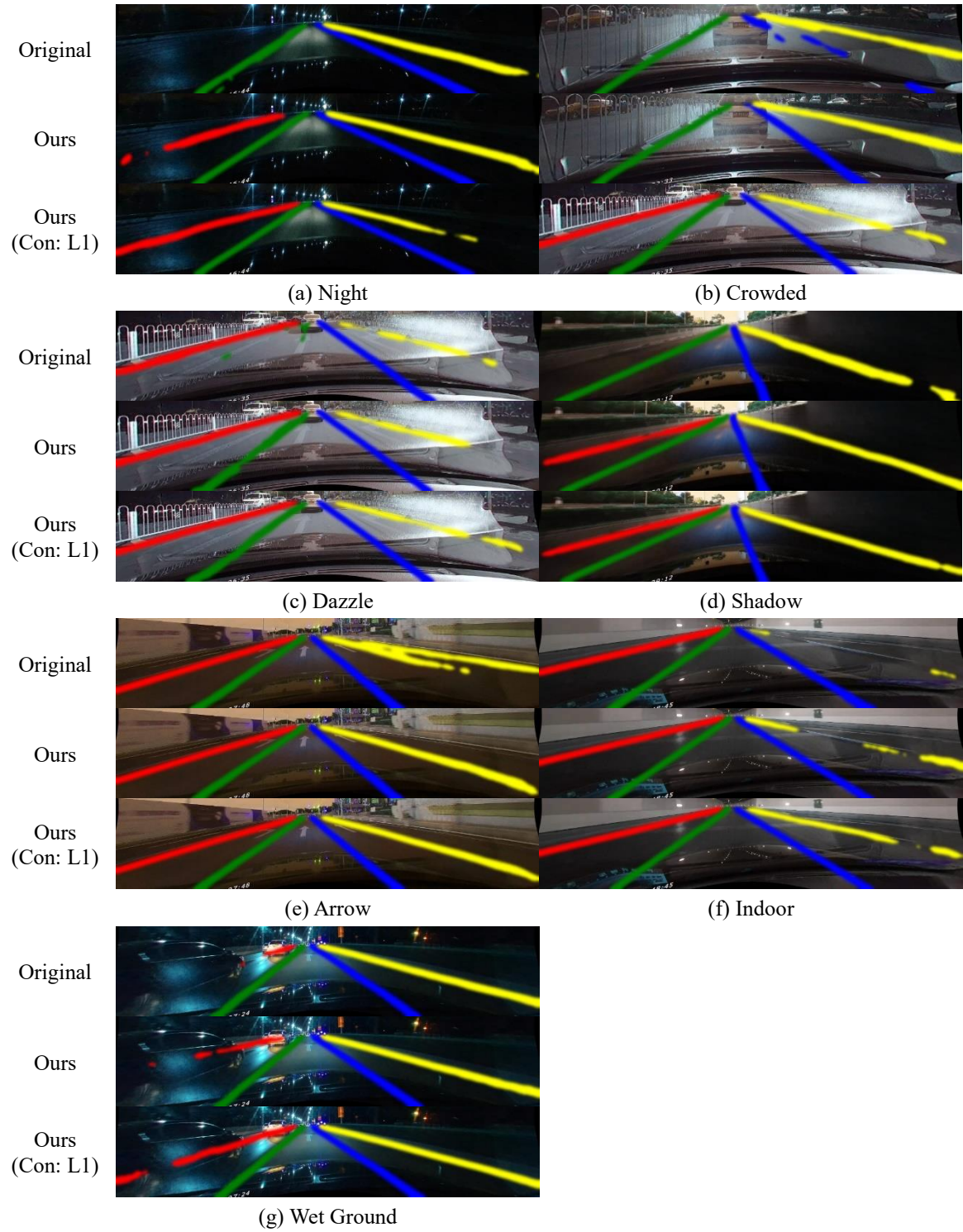


Figure 7. Visualization results of challenging scenes in the CULane test dataset

In this section, we visualize the performance of lane detection in challenging scenarios and compare the difference between our proposed method and the original model. Our

visualization results in Figure 7 clearly show that our method significantly outperforms the original model in complex scenarios such as nighttime, shadowed areas, and indoors. Our method can recognize lane lines more accurately in these challenging scenarios, ensuring drivers can get reliable lane guidance in any environment. It is worth noting that our method performs well both with and without consistency loss. Compared to the original model, our method already achieves significant performance gains without loss of consistency. When we introduce consistency loss, it further strengthens the stability and accuracy of our method, enabling it to maintain excellent performance in different scenarios. In conclusion, our study demonstrates the critical importance of lane detection in challenging scenarios and demonstrates the superiority of our proposed method over the original models. Regardless of applying the consistency loss or not, our method achieves superior performance in various scenarios, which will have a profound impact on improving the safety and reliability of self-driving and assisted-driving systems.

Chapter 5 Conclusion and Future Work

This thesis proposes a feature extraction enhancement strategy based on style translation to solve the problem that the general lane detection model has poor detection performance in challenging scenes due to the lack of diversity in the dataset. We propose to use CycleGAN for image style transfer to enhance the detection performance and robustness of lane detection models at night or in other challenging scenarios. We take advantage of CycleGAN advantage of not requiring paired images to train a domain adaptation model using day and night images from existing lane datasets. We propose a two-stage approach to improve lane detection performance using a feature fusion architecture combined with a domain adaptation model and an image segmentation model. The proposed consistency loss function is used to reduce the difference in the detection ability of the model under different driving scenarios so that the model can perform better. In the test results of the CULane test dataset, the total F1 score of the original ERFNet model is 73.1, and the F1 score of the night is 67.1. After using our method, the performance of the model improved to an F1 score of 74.3 and an F1 score of 69.1 at night. In other challenging scenarios, our method also improves the original model by 0.7% to 15.1%. Experimental results demonstrate the potential of our method to improve lane detection performance in challenging driving scenarios, thereby enhancing driving safety.

In future work, we mainly have several directions to expand our research. In the part of data pre-processing, we can consider removing the part of the car or building in the background first, so that the model can focus on lane detection. In the method part, first, we hope to try more different strategies to train our domain adaptation model. In our preliminary experiments and internal experiments, we mentioned that we tried different strategies to train the domain adaptation model, but we could not achieve the desired experimental results, and

we think this is still a potential research direction. Therefore, we believe we can try more different strategies to train better domain adaptation models to improve the performance of lane detection models. We also want to be able to train domain adaptation models based on images of different styles or based on different brightness. Our current approach mainly trains domain adaptation models based on daytime and nighttime images, achieving good performance in challenging scenarios under different lighting conditions. However, due to the lack of datasets, we were unable to train or test our models for scenes such as rain and snow, nor could we know whether there is a significant difference in the ability of the model under different brightness. In future work, we hope to obtain related datasets to train more domain adaptation models with different styles or different brightness. Finally, in the experimental part, we also hope to use different datasets or use driving recorders to collect new data sets to verify the performance of our method.

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